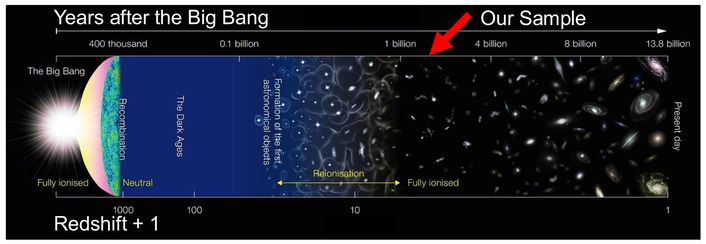
Probing Early Supermassive Black Hole Growth: C IV-Based Mass Estimates in z ~ 4.8 Quasars

Nayera Abdessalam, Richard Green, Weizhe Liu

Steward Observatory, University of Arizona, 933 E. Cherry Ave., Tucson, AZ 85721, USA

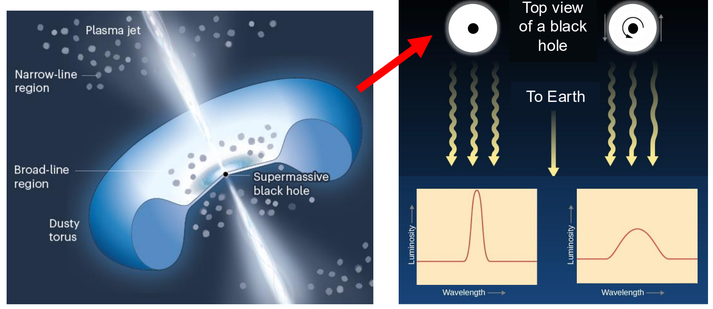
FRANKE HONORS PINNACLE

Background



Schematic timeline of the universe, highlighting early structure formation and the emergence of quasars powered by rapidly growing supermassive black holes.

Quasars are powered by supermassive black holes (SMBHs) with masses ranging from about 10^6 to 10^{10} solar masses. Observations of nearby galaxies show strong correlations between SMBH mass and host galaxy properties, suggesting that black holes and galaxies grow together [9,11,12]. **Understanding how these massive black holes formed so early in the universe remains a major challenge. Measuring their masses is therefore essential for studying both black hole growth and galaxy evolution.**



Emission lines in quasar spectra arise from fast-moving gas near the SMBH, linking observed light to black hole mass measurements.

Mass measurements of the SMBH are made using a spectrum of the quasar light. Emission lines are broadened due to the Doppler effect from high-velocity gas orbiting the SMBH. The velocity reflects the gravitational influence of the SMBH, which can be directly related to its mass. The width of the spectral emission lines trace the velocity of the gas and dust rotating around the SMBH. Known quasar emission lines like Hβ and C IV happen due to gas excitation in the broad-line region of the quasar, with higher-ionization lines originating closer to the black hole.

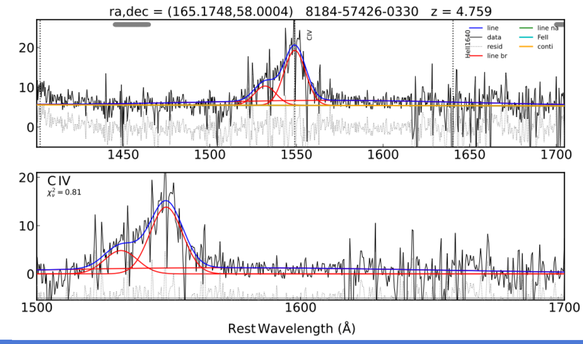
Objectives & Methods

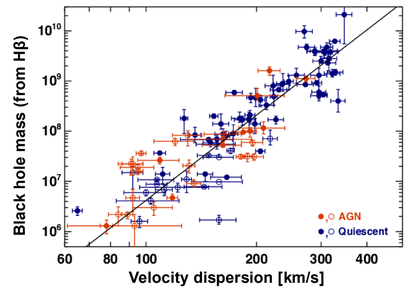
Objectives

1. Compare mass measurements of SMBHs made from line widths of triply ionized Carbon (C IV) versus excited neutral Hydrogen (Hβ)
2. Calibrate C IV measurements with known correction techniques

Sample

14 Quasar Spectra, z ~ 4.8 with C IV data from Sloan Digital Sky Survey & Hβ data from James Webb Space Telescope (JWST)





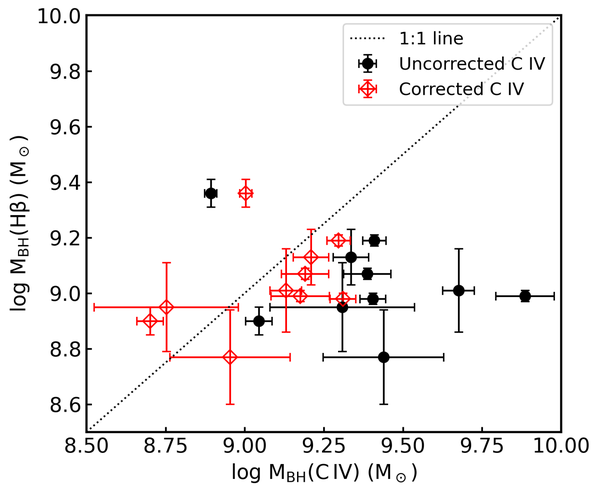
Established Hβ-based BH mass calibration becomes unavailable at high redshift, motivating the use of C IV as a mass estimator.

Line Fitting

- Each spectrum was fit with four model fits with varying broad and narrow line components
- Best fit selected with Bayesian Information Criterion (BIC)
- Extracted FWHM, blueshift, other parameters

Key Results

- The C IV-based black hole mass estimates show a general positive correlation with the comparison mass indicator, indicating that the method captures the overall scaling of SMBH mass.
- There is noticeable scatter around the 1:1 relation, suggesting that additional physical effects (e.g., outflows or line asymmetries) introduce uncertainty into C IV-based mass estimates.
- Systematic offsets from the 1:1 line indicate that uncorrected C IV measurements may bias mass estimates in a consistent direction across part of the sample.
- The spread in the data increases at higher masses, implying that uncertainties or biases in C IV-based estimates may become more significant for the most massive quasars.



Conclusions

Conclusion

Calibrated C IV mass estimates are successful at tracing the gravitational motion of quasar. With continual correction efforts they can be a reliable tool for studying black hole growth in the early universe, particularly in regimes where Hβ cannot be observed. This approach enables consistent SMBH mass measurements for large samples of high-redshift quasars.

Future Work

Future work will expand the sample size and refine the correction calibrations using larger datasets, improving the accuracy of black hole mass estimates across cosmic time. With recent JWST and SPHEREx data, and upcoming data from telescopes like Nancy Grace Roman, we will continue pushing these techniques to increasingly higher redshifts.



(Left) James Webb Space Telescope. (Right) Nancy Grace Roman Telescope, to be launched in late 2026.

Acknowledgements

Full Reference List

Acknowledgements